

“Unless the demand crisis is addressed, the supply-side push will end up in stranded assets, erosion in investor confidence, and failure to export”

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India's pilot hydrogen train running between Jind and Sonipat since July 17, 2026

HIGH COST

The easiest transition is from grey hydrogen used in refineries and fertiliser plants (ammonia) to green hydrogen. India produces 6.5 MT of grey hydrogen, which is exclusively for captive use.

Balashanmugam says transitioning from grey to green hydrogen will double production costs for refineries and increase them by 1–1.5 times for fertiliser plants. “The switch is economically unviable for the industry today. For non-traditional sectors of road transport and shipping, the challenges are creating supporting infrastructure like fuelling facilities and engine modifications. Engines may require small or substantial modifications depending on technology (direct hydrogen burning or fuel-cell based),” he says.

However, the good news is that the NGHM push has lowered the cost of green hydrogen.

From $5-6 per kg before the mission was launched in 2023, the cost of production has fallen to $3.5-5 per kg (without subsidy) and $3-3.75 (with subsidy) in recent tenders. Even at this level, the cost is far higher than the baseline price of $1.9-2.5 per kg for grey hydrogen, according to CRF data.

Besides, the current manufacturing capacity is too low to support a meaningful transition.

According to GH2 India, the operational capacity for electrolysers is 65-70 MW—against the announced capacity of 25 GW and incentivised capacity of 3 GW. For green hydrogen, it is 12,000 tonnes—against the announced capacity of 11.2 MT and incentivised capacity of 862,000 tonnes.

Though such capacities have been created, actual production has not begun. L&T’s electrolyser plant mentioned earlier hasn’t begun production despite receiving a supply order of 100 MW from Indian Oil’s Panipat plant. Balashanmugam says the entire SIGHT incentive of ₹17,490 crore is unclaimed. This subsidy is payable after production as is the case with production-linked incentive schemes.

EXPORTS CONSTRAINTS

In its “demand creation” segment, the NGHM gives greater emphasis to export markets than domestic demand. It says global demand for green hydrogen and